

Reaction kinetics

A Level Chemistry

Rate of reaction

The **rate of reaction** 反应速率 is how fast reactants turn into products. We measure it as the change in concentration (or amount) in each unit of time.

To react, particles must **collide** 碰撞. The **collision frequency** 碰撞频率 is how often the particles hit each other. But not every collision leads to a reaction:

- an **effective collision** 有效碰撞 has enough energy *and* the correct direction, so a reaction happens.
- a **non-effective collision** 无效碰撞 does not have enough energy, or the particles hit at the wrong angle, so nothing happens.

So the rate depends on the **frequency of effective collisions** —how many useful collisions happen each second.

Concentration and pressure

If you increase the concentration of a solution (or the pressure of a gas), the particles are packed closer together. They collide more often, so there are more effective collisions each second, and the rate goes up.

You can calculate a rate from experimental data —for example, the volume of gas made divided by the time taken.

Temperature and activation energy

The **activation energy** 活化能 (E_A) is the minimum energy a collision needs in order to be effective.

The **Boltzmann distribution** 玻尔兹曼分布 is a graph showing how the energies of the molecules are spread out at one temperature. The curve starts at the origin, rises to a peak, then falls away in a long tail. The total area under the curve is the total number of molecules. Only the molecules to the right of E_A have enough energy to react.

When you raise the temperature:

- the curve flattens and spreads to the right, so a **much larger fraction** of molecules now have energy greater than E_A .
- the molecules also move faster and collide more often.

The first effect is the bigger one. This is why a small rise in temperature gives a large rise in rate.

Catalysts

A **catalyst** 催化剂 speeds up a reaction but is not used up itself. **Catalysis** 催化作用 is the name for this action.

A catalyst works by giving the reaction a different **reaction mechanism** 反应机理—a new route with a **lower** activation energy. On the Boltzmann distribution, lowering E_A moves the

line to the left, so more molecules now have enough energy. This means more effective collisions each second, and a faster rate.

On a **reaction pathway diagram** 反应路径图, the catalysed route has a lower energy "hill". The enthalpy change of the reaction, ΔH , is **not** changed by the catalyst.

There are two types:

- a **homogeneous catalyst** 均相催化剂 is in the **same** physical state as the reactants—for example, an acid catalyst dissolved in a solution of liquids.
- a **heterogeneous catalyst** 多相催化剂 is in a **different** state from the reactants—for example, solid iron speeding up the reaction of gases in the Haber process.